

ERC Really Credible RA Test

Selected Exercises

NIKIEMA Pengd-Wende Richard

PhD Candidate in Development Economics

LEO-UCA, Université Clermont Auvergne

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[Replication materials and GitHub repository](#)



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Introduction

This document presents my responses to four selected exercises from the ERC Really Credible RA test. The exercises cover complementary skills expected from a research assistant working on econometrics and causal inference: theoretical asymptotic reasoning, empirical implementation in Stata, software-oriented econometric reasoning, and user support for the `ChaisemartinPackages` ecosystem.

The document is organized as follows. Exercise 1 studies a super-consistent estimator in a uniform model and illustrates the result using a Monte Carlo simulation in Stata. Exercise 3 implements `did_multiplegt_dyn` to revisit the empirical framework of Burgess et al. (2015) on ethnic favoritism in road allocation in Kenya. Exercise 4 discusses how clustered bootstrap standard errors could be computed more efficiently using random weighting vectors, based on an AI-assisted reasoning process. Exercise 6 answers practical user questions related to `did_multiplegt_dyn`.

The Stata code used for the Monte Carlo simulation in Exercise 1 and for the empirical implementation in Exercise 3 is provided in the accompanying do-file.

1 Exercise 1: A Super-Consistent Estimator

Assume that $Y_1, \dots, Y_n$ are i.i.d. random variables uniformly distributed on $[0, \theta]$, where $\theta > 0$ is unknown. The objective is to compare two estimators of θ : a method-of-moments estimator and an estimator based on the sample maximum.

1.1 Method-of-moments estimator

Since $Y \sim \mathcal{U}[0, \theta]$,

$$E(Y) = \frac{\theta}{2}.$$

Replacing the theoretical expectation by the sample mean $\bar{Y}$ yields the method-of-moments estimator:

$$\hat{\theta}_{MM} = 2\bar{Y}.$$

Using the central limit theorem,

$$\sqrt{n}(\bar{Y} - E(Y)) \xrightarrow{d} \mathcal{N}(0, V(Y)).$$

Because

$$\hat{\theta}_{MM} - \theta = 2(\bar{Y} - E(Y)),$$

it follows that

$$\sqrt{n}(\hat{\theta}_{MM} - \theta) \xrightarrow{d} \mathcal{N}(0, 4V(Y)).$$

For a uniform distribution,

$$V(Y) = \frac{\theta^2}{12},$$

so the asymptotic variance becomes

$$4V(Y) = \frac{\theta^2}{3}.$$

Hence, the method-of-moments estimator converges at the standard $\sqrt{n}$ rate.

1.2 Maximum-order-statistic estimator

Consider now the estimator

$$\hat{\theta}_{ML} = \max_{1 \leq i \leq n} Y_i.$$

This estimator is particularly natural because θ determines the upper bound of the support of the distribution. The largest observed realization therefore contains direct information about the unknown endpoint.

To derive its distribution, note that for any $x \in [0, \theta]$,

$$P(\hat{\theta}_{ML} \leq x) = P(Y_1 \leq x, \dots, Y_n \leq x).$$

Using independence,

$$P(\hat{\theta}_{ML} \leq x) = [P(Y \leq x)]^n.$$

Since $Y \sim \mathcal{U}[0, \theta]$,

$$P(Y \leq x) = \frac{x}{\theta},$$

which implies

$$P(\hat{\theta}_{ML} \leq x) = \left(\frac{x}{\theta}\right)^n.$$

1.3 Limiting distribution

Define

$$Z_n = n \left(\frac{\theta - \hat{\theta}_{ML}}{\theta} \right).$$

For $z \geq 0$,

$$\begin{aligned} P(Z_n \leq z) &= P\left(\hat{\theta}_{ML} \geq \theta \left(1 - \frac{z}{n}\right)\right) \\ &= 1 - P\left(\hat{\theta}_{ML} < \theta \left(1 - \frac{z}{n}\right)\right). \end{aligned}$$

Using the previous result,

$$P\left(\hat{\theta}_{ML} \leq \theta \left(1 - \frac{z}{n}\right)\right) = \left(1 - \frac{z}{n}\right)^n.$$

Therefore,

$$P(Z_n \leq z) = 1 - \left(1 - \frac{z}{n}\right)^n.$$

As $n \rightarrow \infty$,

$$\left(1 - \frac{z}{n}\right)^n \rightarrow e^{-z},$$

so

$$P(Z_n \leq z) \rightarrow 1 - e^{-z}.$$

This is the cumulative distribution function of an exponential distribution with parameter 1. Consequently,

$$n \left(\frac{\theta - \hat{\theta}_{ML}}{\theta} \right) \xrightarrow{d} \text{Exp}(1).$$

1.4 Comparison of convergence rates

The two estimators converge at different speeds.

The method-of-moments estimator satisfies

$$\hat{\theta}_{MM} - \theta = O_p \left(\frac{1}{\sqrt{n}} \right),$$

which corresponds to the standard convergence rate in econometrics.

By contrast, the previous result implies

$$\hat{\theta}_{ML} - \theta = O_p \left(\frac{1}{n} \right).$$

Since $1/n$ converges to zero faster than $1/\sqrt{n}$, the maximum-order-statistic estimator reaches the true parameter value at a superior rate.

For this reason,

$$\hat{\theta}_{ML} = \max_{1 \leq i \leq n} Y_i$$

is said to be *super-consistent*. More generally, an estimator is called super-consistent whenever it converges faster than the usual $\sqrt{n}$ rate. This type of behavior frequently arises when the parameter of interest defines the boundary of the support of the distribution, as in the present exercise.

1.5 Monte Carlo illustration

In this simulation,

$$Y_i \sim \mathcal{U}[0, 1],$$

so the true value of the parameter is

$$\theta = 1.$$

The Monte Carlo simulation is implemented in the accompanying Stata do-file as follows:

```
clear all
set seed 12345

program define mc_uniform, rclass
```

```

drop _all
set obs 1000

gen y = uniform()

summarize y, meanonly
return scalar theta_MM = 2*r(mean)
return scalar theta_ML = r(max)

end

simulate theta_MM=r(theta_MM) ///
         theta_ML=r(theta_ML), ///
         reps(1000): mc_uniform

gen err_MM = abs(theta_MM - 1)
gen err_ML = abs(theta_ML - 1)

summarize err_MM err_ML

```

Across 1,000 simulations, the average absolute error of the method-of-moments estimator is approximately

0.0147,

while the average absolute error of the maximum estimator is approximately

0.0010.

Table 1: Monte Carlo comparison of estimators

Estimator	Mean absolute error	Standard deviation
Method-of-moments estimator	0.0147	0.0110
Maximum estimator	0.0010	0.0010

The Monte Carlo evidence is consistent with the theoretical result: $\hat{\theta}_{ML}$ is much closer to the true value $\theta = 1$ than $\hat{\theta}_{MM}$, reflecting its faster convergence rate.

1.6 Asymptotic confidence interval

Let t_q denote the q -th quantile of an exponential distribution with parameter 1:

$$t_q = F^{-1}(q).$$

From the previous result,

$$n \left(\frac{\theta - \hat{\theta}_{ML}}{\theta} \right) \xrightarrow{d} U, \quad U \sim \text{Exp}(1).$$

Thus,

$$P(U \leq t_{1-\alpha}) = 1 - \alpha.$$

Therefore,

$$P \left[n \left(\frac{\theta - \hat{\theta}_{ML}}{\theta} \right) \leq t_{1-\alpha} \right] \rightarrow 1 - \alpha.$$

Equivalently,

$$P \left[\theta \leq \frac{\hat{\theta}_{ML}}{1 - \frac{t_{1-\alpha}}{n}} \right] \rightarrow 1 - \alpha.$$

For large n ,

$$\frac{1}{1 - \frac{t_{1-\alpha}}{n}} = 1 + \frac{t_{1-\alpha}}{n} + o\left(\frac{1}{n}\right).$$

Using Slutsky's lemma and $\hat{\theta}_{ML} \xrightarrow{p} \theta$, the interval

$$IC(\alpha) = \left[\hat{\theta}_{ML}, \hat{\theta}_{ML} + \hat{\theta}_{ML} \frac{t_{1-\alpha}}{n} \right]$$

has asymptotic coverage $1 - \alpha$. This interval is one-sided in nature because $\hat{\theta}_{ML} \leq \theta$ by construction.

1.7 Real-world interpretation

This exercise shows how the upper endpoint of a distribution can be estimated from observed samples. The World War II tank example illustrates this logic clearly. If serial numbers are approximately uniformly distributed between 1 and θ , where θ is the total number of tanks produced in a month, then the largest observed serial number is highly informative about θ .

In the example, the maximum observed serial number was 255, while the true production was 256. This was much more accurate than conventional intelligence estimates. The same reasoning can be applied to other settings where units are sequentially numbered and only a random sample is observed.

2 Exercise 3: Dynamic Difference-in-Differences Analysis

2.1 Objective

This exercise replicates and extends the empirical framework developed by Burgess et al. (2015), who study ethnic favoritism in road allocation across Kenyan districts between 1963 and 2011.

The original paper documents that districts sharing the ethnicity of the president receive disproportionately higher road expenditures during autocratic periods, while this favoritism largely disappears under democracy.

Using the `did_multiplegt_dyn` estimator developed by De Chaisemartin and d'Haultfoeuille (2024), I estimate dynamic treatment effects associated with presidential coethnicity. Compared to traditional two-way fixed effects regressions, this approach allows for a more flexible analysis of treatment timing, treatment dynamics, and placebo-based validation of identification assumptions.

2.2 Data and empirical setup

The dataset is a balanced district-year panel covering 41 Kenyan districts from 1963 to 2011, yielding 2,009 observations.

The outcome variable, `exp_dens_share`, measures the district share of national road development expenditure divided by the district population share in 1962. Following Burgess et al. (2015), this normalization allows expenditures to be interpreted relative to district population size.

The treatment variable, `president`, equals one when a district is coethnic with the serving president in a given year, and zero otherwise.

The Stata do-file first declares the panel structure as follows:

```
xtset distnum year
```

The panel is strongly balanced, with 41 districts observed each year between 1963 and 2011. Approximately 15.9% of district-year observations are treated, while around 31.7% of observations belong to districts that are ever treated during the sample period.

2.3 Connection with Burgess et al. (2015)

The empirical strategy closely follows Burgess et al. (2015). In the original paper, the authors estimate the relationship between presidential coethnicity and road allocation using district and year fixed effects regressions.

My implementation reproduces the core logic of the paper using a modern dynamic Difference-in-Differences framework based on event-study estimation. This framework allows treatment effects to evolve over time while simultaneously testing pre-treatment parallel trends assumptions using placebo estimates.

2.4 Baseline dynamic specification

The benchmark specification, implemented in the accompanying do-file, is:

```
did_multiplengt_dyn exp_dens_share distnum year president, ///  
    effects(5) ///  
    placebo(5) ///  
    cluster(distnum) ///  
    graph_off
```

Five post-treatment dynamic effects and five placebo effects are estimated. Standard errors are clustered at the district level to account for serial correlation within districts over time.

2.5 Baseline results

Table 2: Baseline dynamic treatment effects

Effect	Estimate	Std. Error	Lower CI	Upper CI
Effect 1	3.175	2.181	-1.100	7.450
Effect 2	2.853	1.648	-0.378	6.083
Effect 3	2.423	1.395	-0.311	5.158
Effect 4	3.881	1.009	1.903	5.859
Effect 5	2.739	0.469	1.819	3.658

The estimated post-treatment effects are positive across all horizons. Effects become statistically significant from the fourth post-treatment period onward. The joint nullity test strongly rejects the hypothesis that all treatment effects are equal to zero.

The average cumulative treatment effect equals 3.014, suggesting economically meaningful favoritism toward districts sharing the ethnicity of the president. These findings are broadly consistent with Burgess et al. (2015), who document substantial ethnic favoritism in Kenyan road allocation.

2.6 Placebo tests

Table 3: Placebo estimates

Placebo	Estimate	Std. Error	Lower CI	Upper CI
Placebo 1	0.251	0.320	-0.376	0.879
Placebo 2	0.040	0.519	-0.977	1.057
Placebo 3	0.590	0.691	-0.765	1.945
Placebo 4	0.581	0.923	-1.228	2.391
Placebo 5	0.617	0.977	-1.299	2.532

The placebo estimates are statistically insignificant individually. The joint placebo test yields a p-value of 0.096, which does not provide strong evidence against the parallel trends and no-anticipation assumptions.

2.7 Event-study graphs

The do-file generates two event-study specifications: a baseline specification with five post-treatment and placebo effects, and an extended specification including seven post-treatment and placebo effects.

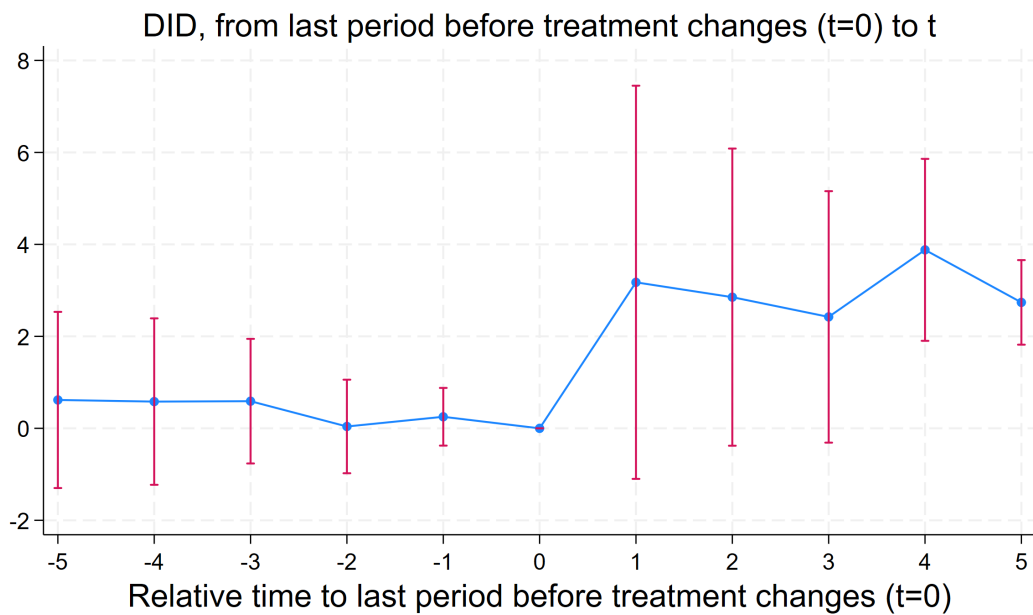


Figure 1: Dynamic effects of presidential coethnicity on road expenditure

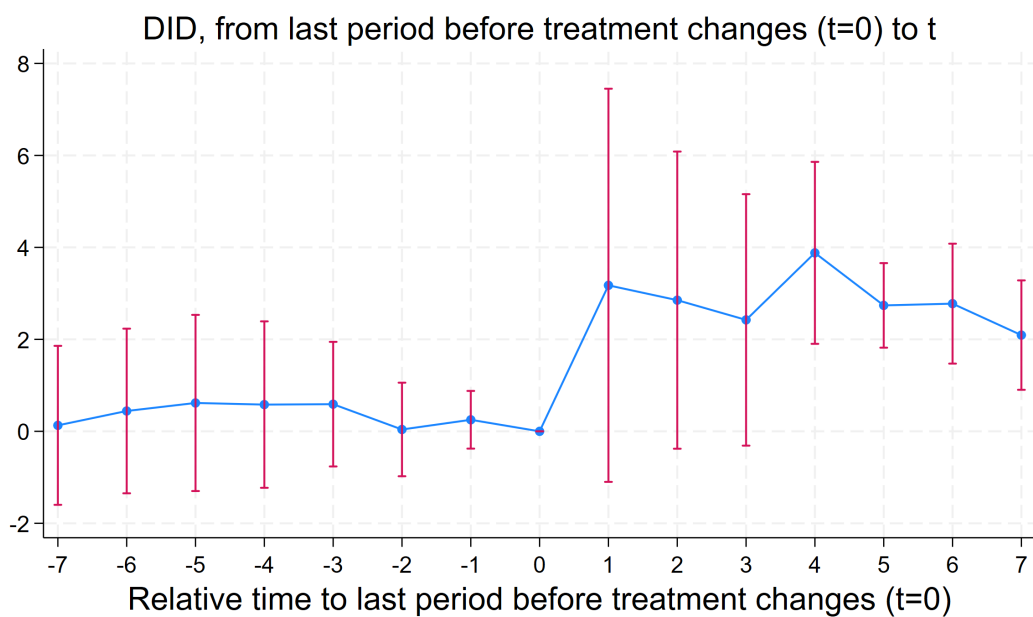


Figure 2: Dynamic effects of presidential coethnicity on road expenditure: longer window

Visual inspection of the event-study graphs (Figures 1 and 2) plays an important role in assessing the validity of the identification strategy. The estimated placebo coefficients corresponding to pre-treatment periods remain statistically insignificant and are generally centered around zero. This pattern suggests that districts that eventually became coethnic with the president and those that did not exhibited similar trends prior to treatment.

These results provide empirical support for the parallel trends assumption. Consequently, the positive and statistically significant post-treatment effects can reasonably be interpreted as evidence of a causal impact of presidential coethnicity on road expenditure allocation.

2.8 Robustness checks

Following Burgess et al. (2015), I introduce baseline district characteristics interacted with a linear time trend. The Stata do-file constructs the linear trend as:

```
gen trend = year - 1963
```

and then generates the following interactions:

```
gen pop1962_trend      = pop1962 * trend
gen area_trend         = area * trend
gen urbrate1962_trend  = urbrate1962 * trend
gen earnings_trend     = earnings * trend
gen wage_employment_trend = wage_employment * trend
gen value_cashcrops_trend = value_cashcrops * trend
```

Two robustness specifications are estimated. The first includes demographic controls interacted with the time trend. The second adds economic controls.

Table 4: Robustness across specifications

Specification	Estimate	Std. Error	Placebo p-value
Baseline	3.014	1.171	0.096
Demographic controls	3.030	1.157	0.099
Full controls	3.108	1.136	0.104

The estimated treatment effects remain highly stable across specifications. Adding demographic and economic controls only marginally changes the estimated cumulative effect. The placebo p-values also remain close across specifications, suggesting that the results are not driven by differential district trends associated with baseline demographic or economic characteristics.

2.9 Switching out of coethnicity

The `did_multiplegt_dyn` estimator primarily relies on treatment transitions to identify causal effects. In the baseline specification, the identification mainly comes from districts switching into coethnicity, that is, districts moving from untreated to treated status.

I additionally investigate whether switching-out effects can be estimated using the `switchers(out)` option. The do-file first identifies treatment transitions:

```
bysort distnum (year): gen delta_president = president - president[_n-1] if _n > 1
tab delta_president

bysort distnum (year): gen switch_out = president[_n-1] == 1 & president == 0 if _n > 1
tab switch_out

bysort distnum: egen n_switch_out = total(switch_out)
tab n_switch_out
```

The data contain 13 transitions into treatment and 13 transitions out of treatment.

I then estimate:

```
did_multiplegt_dyn exp_dens_share distnum year president, ///
    effects(5) ///
    placebo(5) ///
    cluster(distnum) ///
    switchers(out) ///
    graph_off
```

However, the command reports that **no treatment effect can be estimated because Design Restriction 1 is not satisfied**. This indicates that although treatment exits exist in the data, the timing structure of treatment transitions does not satisfy the identification requirements necessary to estimate switching-out effects.

Substantively, investigating treatment exits is informative because it allows us to assess whether the effects of ethnic favoritism are symmetric over time. If favoritism were purely contemporaneous, one would expect road expenditure to decline once a district loses its coethnic connection to the president. However, infrastructure investment is characterized by substantial administrative, financial, and physical inertia. Road projects initiated during periods of coethnic favoritism may continue even after the political connection disappears. As a result, treatment-entry effects may be easier to identify empirically than treatment-exit effects.

From an econometric perspective, the `switchers(out)` specification requires valid comparison groups consisting of units that remain stably treated while other districts switch out of treatment. In this dataset, the relatively small number of treatment exits and the limited persistence of treatment sequences likely prevent the estimator from constructing the counterfactual comparisons required for identification.

More specifically, two issues may explain the failure of the design restriction:

- the number of districts remaining stably treated over sufficiently long horizons may be too small to serve as valid controls;
- treatment sequences may be too short or fragmented to estimate five dynamic post-exit effects while preserving valid comparison groups.

Therefore, while the baseline results clearly document the emergence of ethnic favoritism following treatment entry, the available treatment structure does not allow the separate identification of switching-out effects in this empirical setting.

2.10 Exercise 3 conclusion

This exercise revisits the political economy framework developed by Burgess et al. (2015) using a modern dynamic Difference-in-Differences estimator. The results provide evidence of positive and persistent effects of presidential coethnicity on road expenditure allocation. The robustness checks support the stability of the findings, while the switching-out exercise illustrates an important limitation of the estimator in this empirical setting.

3 Exercise 4: Bootstrap Standard Errors Using Random Weighting Vectors

3.1 Initial discussion with the AI assistant

At first, I was trying to understand whether a clustered bootstrap could be implemented more efficiently in Stata. I therefore started by asking the AI assistant the following question:

“I’m currently doing an econometric analysis in Stata and want to use bootstrapping to test the robustness of my estimates. But I’m facing a big problem: it’s very time-consuming. It seems the resampling method can be replaced by a reweighting approach to reduce the time required for this process. What do you think?”

The AI assistant explained that a clustered bootstrap can indeed be reformulated as a weighting problem. Instead of physically rebuilding a bootstrap sample at each replication, one can keep the original dataset fixed and assign random cluster-level multiplicity weights corresponding to the number of times each cluster would have been selected in the bootstrap sample.

Summary. The key insight from this first exchange was that standard clustered bootstrap resampling can be represented through random cluster-level weights. This can reduce computation time because the dataset no longer needs to be repeatedly copied, resampled, and restored.

3.2 Mathematical formalization

I then asked the AI assistant to formalize the procedure mathematically in the context of `did_multipllegt_dyn`.

“Can you formalize it mathematically in the context of `did_multipllegt_dyn` currently computing standard errors through a clustered bootstrap when treatment is continuous?”

Let $g = 1, \dots, G$ index clusters. In bootstrap replication b , draw G clusters with replacement. Let

$$w_g^{(b)}$$

denote the number of times cluster g is selected. For each observation i belonging to cluster $g(i)$, define

$$w_i^{(b)} = w_{g(i)}^{(b)}.$$

The bootstrap estimator can then be written as

$$\hat{\theta}^{(b)} = \hat{\theta}(w^{(b)}),$$

where the estimator is computed using the bootstrap weights $w^{(b)}$.

The bootstrap variance is

$$\hat{V}_{boot}(\hat{\theta}) = \frac{1}{B-1} \sum_{b=1}^B \left(\hat{\theta}^{(b)} - \bar{\hat{\theta}}^{boot} \right)^2,$$

where

$$\bar{\hat{\theta}}^{boot} = \frac{1}{B} \sum_{b=1}^B \hat{\theta}^{(b)}.$$

Summary. The weighted bootstrap is mathematically equivalent to the standard clustered bootstrap. The difference is computational: the classical bootstrap rebuilds datasets, while the weighted bootstrap keeps the original dataset fixed and updates a random weight vector.

3.3 Prototype Stata routine

A routine could be implemented in Stata.

```
capture program drop weighted_bootstrap_proto
program define weighted_bootstrap_proto, rclass
```

```
    syntax varlist(min=4 max=4) [if] [in], ///
        CCluster(varname) ///
        REPS(integer)
```

```
    marksample touse
```

```
    tokenize `varlist'
    local y `1'
    local group `2'
    local time `3'
    local treatment `4'
```

```
    preserve
    keep if `touse'
```

```
    tempfile base results
    save `base', replace
```

```
    levelsof `cluster', local(clusters)
```

```

local G : word count 'clusters'

postfile handle rep effect1 using 'results', replace

forvalues b = 1/'reps' {

    use 'base', clear
    gen boot_w = 0

    forvalues j = 1/'G' {
        local draw = ceil(runiform()*'G')
        local cl_draw : word 'draw' of 'clusters'
        replace boot_w = boot_w + 1 if 'cluster' == 'cl_draw'
    }

    quietly did_multiplegt_dyn 'y' 'group' 'time' 'treatment' ///
        [pw=boot_w], ///
        effects(1) ///
        graph_off

    capture scalar effect1_b = e(Effect_1)

    if _rc == 0 {
        post handle ('b') (effect1_b)
    }
}

postclose handle

use 'results', clear
summarize effect1

return scalar se_effect1 = r(sd)

restore

end

```

This is only a prototype. A production-level implementation would need to extract all relevant estimands returned by `did_multiplegt_dyn`, including the ATE, dynamic effects, placebo effects, and covariance matrices.

3.4 Implications for controls, trends, and weights

The main implementation challenge is that bootstrap weights must enter all internal steps of the estimator.

If the command includes controls, residualization steps must be weighted. For example, if the estimator residualizes the outcome with respect to controls X , the weighted projection should solve:

$$\min_{\beta} \sum_i w_i^{(b)} (Y_i - X_i' \beta)^2.$$

The same logic applies to linear and parametric trends. If group-specific trends are estimated, they must be estimated using the bootstrap weights:

$$\sum_i w_i^{(b)} (Y_i - \alpha_{g(i)} - \lambda_{g(i)} t_i)^2.$$

If the user already specifies weights a_i , the bootstrap weights should not replace them. Instead, the effective weight should be:

$$\omega_i^{(b)} = a_i \times w_i^{(b)}.$$

3.5 ATE, dynamic effects, and placebos

For each bootstrap replication b , the command should recompute:

$$\widehat{ATE}^{(b)},$$

the dynamic effects,

$$\widehat{\delta}_1^{(b)}, \dots, \widehat{\delta}_K^{(b)},$$

and the placebo effects,

$$\widehat{\pi}_1^{(b)}, \dots, \widehat{\pi}_L^{(b)}.$$

The bootstrap standard errors should then be computed from the empirical dispersion of these replicated estimates.

The main risk is to apply bootstrap weights only at the final aggregation stage while leaving earlier nuisance-parameter steps unweighted. That would not reproduce the classical clustered bootstrap.

3.6 Exercise 4 conclusion

The discussion with the AI assistant clarified that clustered bootstrap resampling can be represented through random cluster-level multiplicity weights. This approach is computationally

attractive because it avoids repeated dataset reconstruction. However, to be valid, the weights must propagate through the full estimation pipeline, including controls, trends, user-specified weights, ATE computation, dynamic effects, and placebo effects. In sum, AI can be a useful tool to make substantial progress in research, especially when clarifying intuitions, exploring alternative formulations, or translating conceptual ideas into code. However, it should be used with caution. One important limitation is that the answers provided by AI may depend strongly on how the question is framed. Therefore, when using AI to clarify an econometric intuition, one should be careful not to introduce confirmation bias through overly leading or restrictive prompts.

4 Exercise 6: User Questions for ChaisemartinPackages

4.1 Question 1: Multiple funding pots

The user has four distinct treatment variables corresponding to four different funding pots allocated in different years. If the objective is to estimate the effect of each funding pot separately, the cleanest approach is to run separate `did_multiplegt_dyn` estimations, using one treatment variable at a time.

For example, in Stata:

```
did_multiplegt_dyn outcome group time pot1_treatment, ///  
    effects(4) placebo(4) cluster(group)
```

```
did_multiplegt_dyn outcome group time pot2_treatment, ///  
    effects(4) placebo(4) cluster(group)
```

```
did_multiplegt_dyn outcome group time pot3_treatment, ///  
    effects(4) placebo(4) cluster(group)
```

```
did_multiplegt_dyn outcome group time pot4_treatment, ///  
    effects(4) placebo(4) cluster(group)
```

This approach estimates the dynamic effect of each funding pot separately. It is important to verify that each treatment variable is well defined at the group-time level and that there is enough variation in treatment timing.

If the four pots are mutually exclusive and conceptually represent a single policy intensity, another possibility is to construct a combined treatment variable. However, this changes the estimand: the command would then estimate the effect of changes in the combined treatment, not the separate effect of each pot.

4.2 Question 2: Number of observations used for the graph

The number of observations used in the event-study graph may differ from the total number of observations in the dataset because the estimator only uses observations that contribute to the identification of the requested dynamic and placebo effects.

Some observations may be excluded because they belong to groups that never change treatment status, do not have enough pre-treatment or post-treatment periods, cannot contribute to the requested event-study horizon, or are excluded by the switching structure required for identification.

For example, increasing the event-study window from

```
effects(5) placebo(5)
```

to

```
effects(10) placebo(10)
```

may reduce the effective estimation sample because fewer groups have sufficiently long histories to contribute to all requested horizons.

4.3 Question 3: Monthly seasonality

The user is working with monthly product-level panel data and would like to account for seasonal variation in prices. This is a legitimate concern because prices may vary systematically across calendar months.

The `controls()` option does not allow Stata factor-variable or time-series operators, such as `i.month` or `i.year`. Using such a specification returns:

factor-variable and time-series operators not allowed.

A practical solution is to generate the dummy variables manually before calling `did_multipllegt_dyn`:

```
tab month, gen(month_fe_)

did_multipllegt_dyn outcome group time treatment, ///
  effects(12) ///
  placebo(12) ///
  controls(month_fe_2-month_fe_12) ///
  cluster(group)
```

One month category is omitted to avoid perfect multicollinearity.

4.4 Question 4: Additional fixed effects and a second policy

The user would like to account for a second policy that occurs after the main treatment and only among treated units. This is important because the second policy may contaminate the estimated dynamic effects of the first treatment.

However, adding a fixed effect for the second policy is not equivalent to controlling for a predetermined covariate. Since the second policy occurs after treatment, it is a post-treatment variable. Controlling for it may therefore change the estimand and absorb part of the dynamic effect of the initial treatment.

A safer first step is to describe the timing of the second policy relative to the main treatment:

```
bysort group (time): gen first_treat_time = time if treatment == 1
bysort group: egen G_main = min(first_treat_time)

bysort group (time): gen first_policy2_time = time if policy2 == 1
```

```
bysort group: egen G_policy2 = min(first_policy2_time)
```

```
gen lag_policy2 = G_policy2 - G_main
```

```
tab lag_policy2
```

If the second policy always occurs after the main treatment, one possible robustness check is to restrict the event-study window to horizons before the second policy occurs, or to estimate the effect among units not yet exposed to the second policy.

More generally, if the objective is to jointly study dynamic effects of several treatments, the user should consider methods specifically designed for multiple treatments.

General Conclusion

This test was valuable because it combined theoretical econometrics, empirical implementation, software reasoning, and applied user support. Exercise 1 highlighted the difference between standard $\sqrt{n}$ -consistent estimators and super-consistent estimators. Exercise 3 showed how modern dynamic Difference-in-Differences estimators can be used to revisit an important political economy result. Exercise 4 encouraged reflection on the computational architecture of econometric software, especially the relationship between clustered bootstrap resampling and random weighting. Exercise 6 emphasized the importance of clear communication when helping users apply advanced econometric tools.

I thank the recruitment team for proposing a test that is both demanding and intellectually stimulating. The exercises were useful not only as an evaluation tool, but also as an opportunity to deepen my understanding of modern Difference-in-Differences methods, computational implementation, and research-assistant tasks in applied econometrics.

References

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